The lecture begins with a recap of childhood mathematical concepts, including vectors, matrices, calculus, and probability, emphasizing their importance in machine learning.

Vectors

Defined as entities representing magnitude and direction (physics) or ordered lists of numbers (mathematics, AI/ML). In data science, vectors represent features and data points in a coordinate system. Row and column vectors are introduced.

- **Example:** A vector (2, 3) can represent a data sample with two features (e.g., study hours and sleep hours).
- **Vector addition** involves adding corresponding components.
- **Scalar multiplication** changes the magnitude but not the direction of a vector.

Matrices

Defined as collections of vectors arranged in rows and columns. Matrices are fundamental in neural networks, image processing, and data manipulation. Matrix dimensions are expressed as row x column (e.g., 2x2).

- **Matrix multiplication** is explained (row into column).
- **Rule:** Matrix multiplication is only possible when the number of columns in the first matrix equals the number of rows in the second matrix. The resulting matrix dimension is (rows of first matrix) x (columns of second matrix).

Dot Product

The dot product of two vectors measures how aligned they are.

- **Directionality:** A positive dot product indicates similarity, while a zero dot product indicates orthogonality (90-degree angle). Negative dot products indicate opposite directions.
- **Cosine similarity**, based on the dot product, provides a value between -1 and 1, indicating the angle between vectors.
- **Use Cases:** Dot products are used in vector databases and recommendation systems.

Eigenvalues and Eigenvectors

Eigenvectors maintain their direction when a linear transformation is applied, only changing in scale (magnitude). Eigenvalues represent the scaling factor.

- **PCA:** Eigenvalues and eigenvectors are used in Principal Component Analysis (PCA) for dimensionality reduction, retaining the meaning of the data while reducing the number of features.
- **Example:** A spring can be compressed or expanded (changing magnitude) without changing its direction.
- **Algorithms:** Eigenvectors are also used in Google's PageRank algorithm.

Differential Calculus

Differentiation is about changes. dy/dx represents the rate of change of y with respect to x.

- **Example:** If $y = 2x$, then $dy/dx = 2$, meaning y changes by 2 units for every 1 unit change in x.
- **Slope:** dy/dx represents the slope (tangent) of a curve. A constant slope indicates a straight line.

Gradient

A multi-variable differentiation. When a variable depends on multiple independent variables (e.g., $z = x^2 + y^2$), partial derivatives are used ($\partial z/\partial x$ and $\partial z/\partial y$).

Chain Rule

Used when dealing with functions of functions. It helps find the rate of change of a composite function.

- **Example:** If $y = f(g(x))$, then $dy/dx = (dy/dg) * (dg/dx)$.
- **Machine Learning:** The chain rule is crucial for understanding **backward propagation** in machine learning.

The lecture concludes by emphasizing the importance of understanding these concepts for the upcoming discussions on linear regression, loss functions, and optimization algorithms. The next classes will focus on how machines learn, backward propagation, and the mathematics behind machine learning algorithms.